Chapter 7: Nonparametric and Categorical Data

Statistics for Environmental Science

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Table of contents

Parametric vs. nonparametric tests **1**

The Wilcoxon signed-rank test **2**

The Mann-Whitney-Wilcoxon test **7**

Categorical data: the chi-square tests **10**
The chi-square goodness-of-fit test 12
The chi-square test of independence 13

Chapter Summary **16**

Parametric vs. nonparametric tests

Chapter 6 covered three forms of the t-test (one-sample, paired, and independent-samples, also called two-sample). All three need a continuous outcome that is roughly normal (for the paired test, the difference scores), and the pooled independent-samples test also assumes similar variance across the two groups. This chapter covers what to do when those assumptions aren't met. We'll also cover what to do if your data aren't continuous at all.

So far, we've been looking at **parametric** tests, without calling them that. Every test needs a known null distribution for its test statistic, since that's the shape we compare our result against to get a p-value. A test is parametric when we can only know that shape by assuming something specific about the population the data came from.

For the t-test, that assumption is normality. The t-distribution isn't the shape of your raw data. It's the shape the t-statistic takes across repeated samples, and we can only derive it because we assumed the population itself was normal, or relied on the central limit theorem for a large enough sample.

When the distributional assumptions are met, parametric tests are generally the most powerful available. However, when distributional assumptions are not met, a parametric test's p-value can be too small or too large, and without simulation it is hard to tell which. The worse the departure from the assumed distribution, the larger the possible error.

Nonparametric tests also have a known null distribution for their test statistics, and they get it without assuming anything about the population's shape. They still make assumptions, and those assumptions are usually easier to meet than normality.

Many nonparametric tests are run on the ranks of the data (first, second, third … last), rather than on the raw data. The median is a familiar example. It is the middle value once the data are put in order.

Rank-based tests also work on ordinal data. Because the Wilcoxon and Mann-Whitney-Wilcoxon tests only need a rank ordering, neither is limited to interval- or ratio-scale measurements the way a t-test is. A t-test needs a mean, which requires numbers where the *distance* between values means something. Plenty of environmental data doesn't work that way: a water-quality rating (poor, fair, good, excellent), a disturbance-severity class, a Likert-style survey response. You can put those in order, but "the average disturbance class was 2.4" isn't meaningful the way "the average pH was 6.8" is.

This chapter covers two different kinds of tools:

- **Nonparametric alternatives (to t-tests).** Your outcome is still continuous, but it's skewed, has outliers, or your sample is too small to trust normality. The **Wilcoxon signed-rank test** (Wilcoxon 1945) and **Mann-Whitney-Wilcoxon test** (Mann and Whitney 1947) answer the same questions as the one-sample/paired and independent-samples t-tests, respectively, without assuming normality.
- **Categorical data.** Your outcome isn't a number at all: a site either has an invasive species or it doesn't, a sample either exceeds a threshold or it doesn't. None of the tests above apply here; you need the **chi-square (χ^2) tests**.

We'll state the assumptions explicitly for each test as we go through them.

The Wilcoxon signed-rank test

Chapter 6 built the paired t-test around this same example, wetland restoration and soil moisture, with a full 20-plot dataset, and drew the design in Figure 6.7. The same plots are measured before and after, so each plot contributes one difference. Let's run the same design on a separate, smaller study of 12 plots, and this time check whether we should trust the result before reporting it. The data in this example are invented.

Table 1: Soil moisture (%) before and after restoration, 12 plots.

Plot	Before	After	Difference
1	20	26	6
2	15	21	6
3	22	21	-1
4	18	22	4
5	16	23	7
6	24	23	-1
7	19	18	-1
8	14	22	8
9	21	21	0
10	17	17	0
11	25	25	0

Plot	Before	After	Difference
12	20	23	3

Step 1: run the paired t-test, the way Chapter 6 taught it.

Table 2: Paired t-test: soil moisture before vs. after restoration.

Mean difference	t	df	p
2.58	2.57	11	0.026

$t(11) = 2.57$, $p = 0.026$, a significant result. Restoration looks like it increased soil moisture.

Step 2: check the assumptions. Let's look at the differences before trusting that p-value.

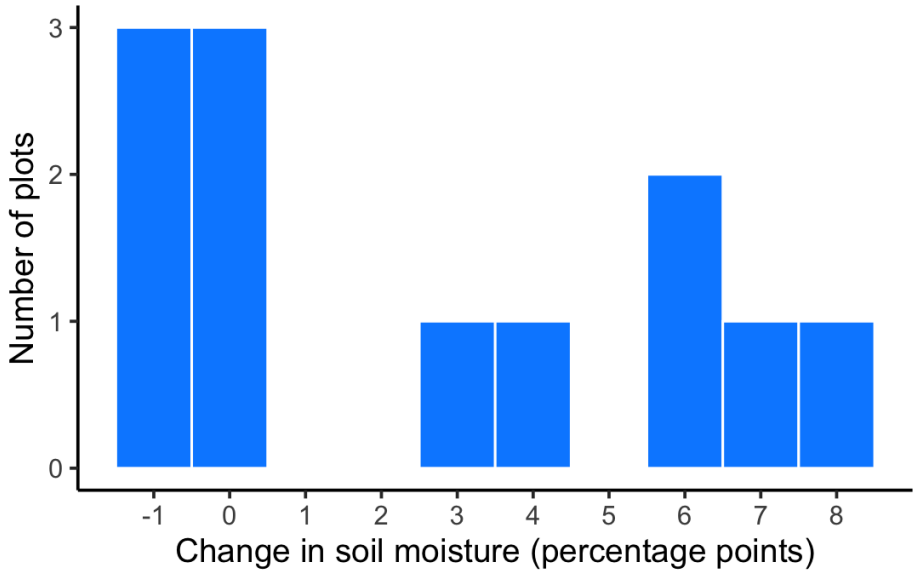


Figure 1: Change in soil moisture per plot. Three plots showed no change at all, and the positive changes stretch out much further than the negative ones: not the symmetric, bell-shaped pattern a paired t-test assumes.

Table 3: Shapiro-Wilk normality test on the differences.

Statistic	p
0.85	0.039

The Shapiro-Wilk test agrees with the picture ($p = 0.039$). Three plots didn't change at all, three dropped slightly, and the rest rose by 3 to 8 points. That is not symmetric, let alone normal. The t-test's "significant" result depends on an assumption the data don't meet, so we turn to the Wilcoxon signed-rank test.

The question. Did soil moisture systematically change after restoration, or is any change we see just plot-to-plot noise?

- H_0: no systematic shift; the differences are symmetric around a median of 0 (positive and negative ranks are balanced).
- H_a: there is a systematic shift; the median difference is not 0.

How it works, conceptually. Instead of using the actual sizes of the differences, the test ranks the *absolute* differences from smallest to largest, then checks whether the positive-signed ranks and negative-signed ranks are roughly balanced. Balanced ranks are the null case, meaning a plot's rank (how extreme its change was) has nothing to do with whether that change was an increase or a decrease. If soil moisture really increased at most plots, the larger ranks should belong mostly to positive differences instead. Because the test uses ranks, the plot that rose by 8 points gets the top rank, 9, and it would get the same rank if it had risen by 30. One extreme plot can't dominate the result the way it could dominate a mean.

Its assumption. The Wilcoxon signed-rank test does not require normality, but it assumes the differences come from a symmetric distribution around their median. That's a milder assumption than normality (symmetric-but-not-normal distributions are common), but our histogram above strains it: the positive side stretches to 8, the negative side barely reaches -1. We'll run the test anyway, since it's still the better-justified choice here, and note the problem when we interpret the result.

```
wilcox.test(differences, mu = 0)
```

Table 4: Wilcoxon signed-rank test on restoration soil-moisture differences.

Statistic (V)	p
39	0.057

The three plots with no change drop out of the test. A zero difference can't be ranked as positive or negative, so the Wilcoxon test runs on the remaining 9 plots instead of all 12. The rank test already ignores how large each change is, and dropping the zeros means it also works with less data than the t-test. The zeros and the tied differences (two plots at $+6$, three at -1) also keep R from using the exact null distribution, so it falls back to a normal approximation.

How V is calculated. Drop the zero differences, take the absolute value of what's left, rank those absolute values from smallest to largest (tied absolute values share the average of their ranks), then sum the ranks belonging to the *positive* differences. That sum is V.

Table 5: Ranking the nonzero restoration differences to calculate V.

Difference	Abs. difference	Rank	Sign
6	6	6.5	+
6	6	6.5	+
-1	1	2.0	-
4	4	5.0	+
7	7	8.0	+
-1	1	2.0	-
-1	1	2.0	-
8	8	9.0	+
3	3	4.0	+

V is the sum of the ranks where Sign is "+": $6.5 + 6.5 + 5 + 8 + 9 + 4 = 39$, matching the table R gave us above.

How the null distribution of V is built. Under H_0, each plot's rank has an independent 50/50 chance of being positive or negative. That gives 2^n equally likely sign patterns for n nonzero differences (here $n = 9$, not 12, since the zeros dropped out). Summing the positive ranks for each of those patterns builds the null distribution of V. It has a known mean and variance:

$$E[V] = \frac{n(n+1)}{4}, \qquad \mathrm{Var}(V) = \frac{n(n+1)(2n+1)}{24}$$

Plugging in our $n = 9$ nonzero plots:

$$E[V] = \frac{9 \times 10}{4} = 22.5, \qquad \mathrm{Var}(V) = \frac{9 \times 10 \times 19}{24} = 71.25$$

That's an SD of about 8.4. Our observed $V = 39$ is about 1.96 SDs above that expectation ($\frac{39-22.5}{8.4} \approx 1.96$).

R's $p = 0.057$ comes from this normal approximation, with two small corrections. Because V moves in steps rather than smoothly, R shrinks the gap between V and 22.5 by 0.5, which raises p. It also adjusts the variance for the tied ranks, which lowers p a little. The first correction is larger, so p ends up slightly above the 0.05 a z of 1.96 would give. For small, tie-free samples with no zeros, R can instead read the exact null distribution directly.

To see the null distribution, we keep our 9 ranks and reassign their signs at random 20,000 times. No new plots are measured. Each repetition gives one V, the value our data would have produced if each plot's change had been equally likely to go up or down. The critical values marked on the figure come from listing all 512 sign patterns. That list is the exact null distribution, and the simulation approximates it.

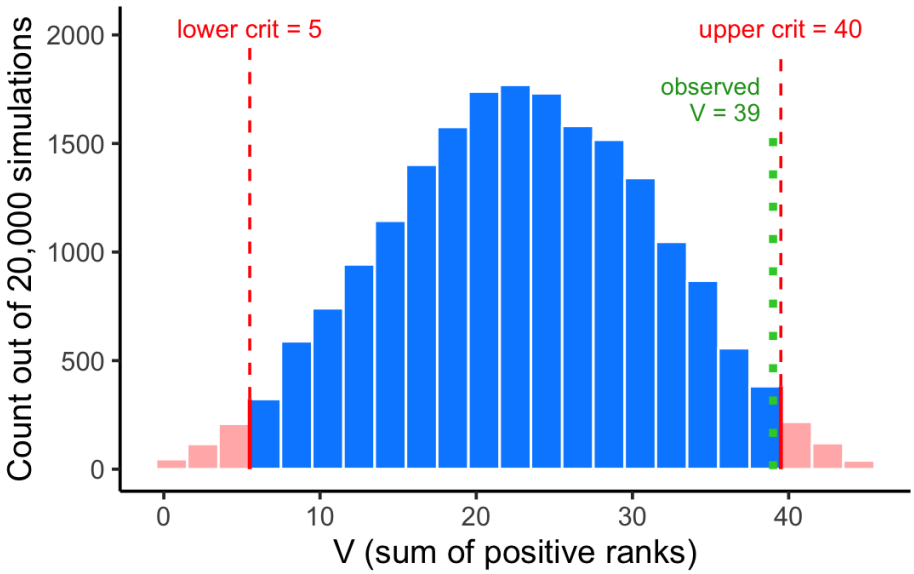


Figure 2: Simulated null distribution of the Wilcoxon signed-rank statistic V, built from 20,000 random sign-flips of the 9 nonzero ranked differences. Shaded regions mark the two-tailed 5% rejection region; our observed V = 39 falls just short of the upper critical value, 40, outside the rejection region.

Our observed $V = 39$ falls just short of the upper critical value, 40, so it is not in the 5% region. V can only take certain values, so the boundary can't fall exactly at 2.5%. A V of 39 or more comes up just over 2.5%

of the time under H_0, which puts the two-tailed p just over 0.05. The result is not significant, though it is close to the cutoff.

The test does not need normality because nothing in this simulation assumed a shape for soil moisture. The only inputs were our 9 ranks and the 50/50 sign rule from H_0.

So far. t-test $p = 0.026$ (significant), Wilcoxon $p = 0.057$ (not quite). The two tests reach different verdicts on the same 12 plots because the t-test's numerator and denominator both used the exact magnitudes the Wilcoxon test deliberately ignores.

The sign test. There's an even simpler nonparametric option, the **sign test**, which looks only at whether each difference is positive or negative. It discards the magnitude entirely and asks whether pluses outnumber minuses more than chance would suggest. The sign test makes no assumption about the distribution of the differences, not even symmetry. Its only assumption is the one every test in this book shares, independent observations from a sound study design (see Chapter 6).

We already flagged the symmetry assumption as shaky for these data, so the sign test avoids the one assumption the Wilcoxon test still needed. That robustness comes at a big cost in power, though, so it's usually a last resort.

The sign test uses the binomial distribution from Chapter 3. `binom.test(6, 9, p = 0.5)` asks how often chance alone would give 6 or more increases out of 9, or 3 or fewer, if each plot were equally likely to go up or go down ($\theta = 0.5$, the binomial success probability from Chapter 3).

```
binom.test(6, 9, p = 0.5)
```

Table 6: Sign test on restoration soil-moisture differences.

Plots increased	Plots total	p
6	9	0.508

Six of the 9 nonzero plots increased, and $p = 0.508$ is far from significant.

The full picture. The three tests reach different verdicts on the same 12 plots, and each uses less of the data than the one before it (Table 7).

Table 7: Three tests on the same 12 restoration plots.

	Paired t-test	Wilcoxon signed-rank	Sign test
What it uses	Full size of each difference	Ranks of the nonzero differences	Direction only
Plots used	12	9	9
Assumption about shape	Differences roughly normal	Differences symmetric around their median	None
Result	$t(11) = 2.57$	$V = 39$	6 of 9 increased
p	0.026	0.057	0.508

That's not an argument that nonparametric tests are worse. The t-test gave the smallest p-value here because it used the full size of every change, including the increases of 3 to 8 points. That p-value is trustworthy only if the differences are roughly normal, and these are not. Check the assumptions before you run the test, so the result cannot influence which test you choose.

The Mann-Whitney-Wilcoxon test

The same check applies to the independent-samples t-test. Many cities are replacing orange sodium streetlights with white LED lights, and ecologists want to know whether the switch changes how much bats use lit areas. Researchers placed acoustic detectors at 8 sites under sodium lights and 8 sites under LED lights and counted bat passes per night, the number of times a bat flew past the detector. The data in this example are invented. Figure 3 gives the design.

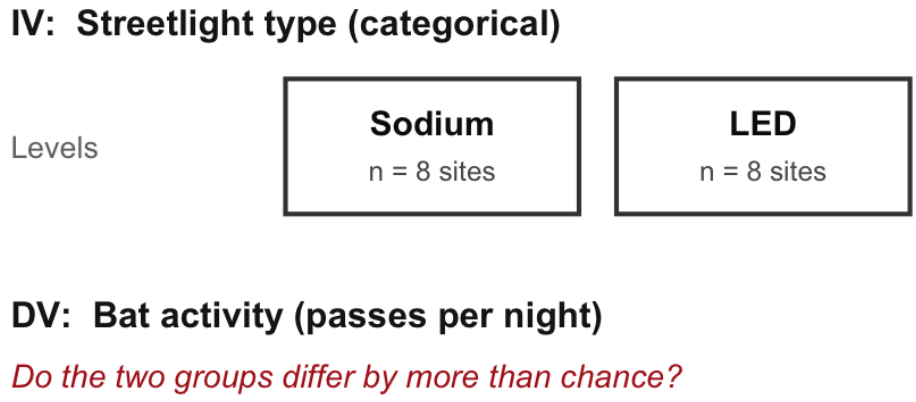


Figure 3: An independent-groups design. Unlike the paired design earlier in this chapter, the two boxes hold different sites, so there is nothing to subtract within a unit.

Table 8: Bat passes per night at sites under sodium and LED streetlights.

Site	Sodium	LED
1	21	17
2	16	18
3	22	18
4	16	12
5	17	11
6	16	11
7	17	11
8	34	10

Step 1: run the t-test, the way Chapter 6 taught it.

Table 9: Independent-samples t-test: sodium vs. LED bat activity.

Mean difference	t	df	p
6.38	2.54	14	0.024

$t(14) = 2.54$, $p = 0.024$, a significant result. Bat activity looks higher under sodium lights.

Step 2: check the assumptions. Chapter 6 said to check them before trusting a p-value, so we look at the data now.

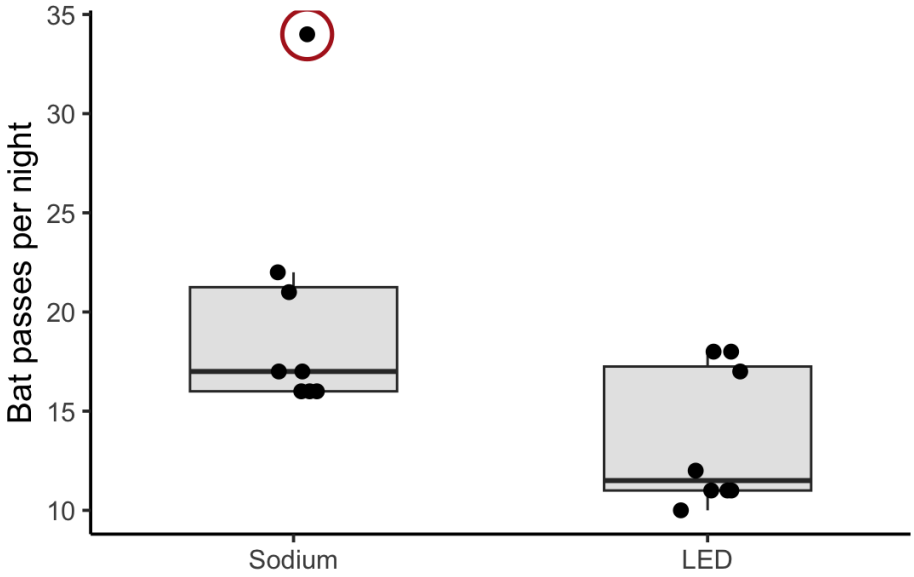


Figure 4: Bat passes per night by streetlight type. The circled sodium site has an unusually high count, visibly pulling that group's distribution away from a normal, bell-shaped one.

The circled site has 34 passes, 12 more than any other site in either group. We don't just throw away outliers, since the site may be near a hedgerow or tree line that bats follow at night. But it does make the sodium group far from normal:

Table 10: Shapiro-Wilk normality test, by group.

Group	Statistic	p
Sodium	0.70	0.002
LED	0.77	0.015

Both groups fail the Shapiro-Wilk test, the sodium group badly ($p = 0.002$). The t-test's "significant" result depends on an assumption the data don't meet. The Mann-Whitney-Wilcoxon test is designed for this situation.

The question. Does bat activity differ between sites under sodium and LED lights, or is the difference we see consistent with random site-to-site variation?

- H_0: no difference in bat activity between sodium and LED sites; ranks are well-mixed between the groups.
- H_a: bat activity tends to rank systematically higher in one group than the other.

How it works, conceptually. Pool all the observations from both groups together, rank them from smallest to largest, then compare the sum of ranks in Group A to the sum of ranks in Group B. If the groups don't differ, their ranks should be well mixed. If one group tends to have higher values, its ranks should be higher too. The outlier site gets the highest rank, 16, whether it had 23 passes or 34. Its size enters the t-test through both the mean and the standard deviation, but the Mann-Whitney-Wilcoxon test uses only its rank.

Its assumption. The Mann-Whitney-Wilcoxon test assumes both groups' parent distributions have the same shape. It does not require normality. Distributions with the same shape also have equal variances, but

equal variances alone do not meet the assumption, because two distributions can share a variance and differ in shape. If the two groups have differently-shaped distributions (say, one skewed and one symmetric), the test still runs, but interpreting a significant result purely as "the medians differ" becomes murkier, since it could be a difference in location, shape, or both.

Our two groups don't quite share a shape. The sodium group is strongly right-skewed, mostly because of the one 34-pass site, while the LED group is only mildly skewed. So read a result from this test as "bat activity tends to be higher in one group," not strictly as a difference in medians.

```
wilcox.test(sodium, led)
```

Table 11: Mann-Whitney-Wilcoxon test: sodium vs. LED bat activity.

Statistic (U)	p
50	0.063

R labels this statistic W; for two groups it is the Mann-Whitney U. R falls back to the normal approximation again, because the pooled counts include ties (three sites each at 11, 16 and 17, and two at 18).

How the null distribution of U is built. The idea is the same as for the Wilcoxon signed-rank test, but the null distribution is built from group membership instead of signs. Under H_0, both groups are draws from the same population. Once you pool and rank all $N = n_A + n_B = 16$ values, chance alone decides which ranks end up in the sodium group, and each of the $\binom{16}{8}$ ways of splitting the 16 ranks into two groups of 8 is equally likely. Those splits make up the null distribution of U, and its mean and variance are known:

$$E[U] = \frac{n_A n_B}{2}, \qquad \mathrm{Var}(U) = \frac{n_A n_B (n_A + n_B + 1)}{12}$$

Plugging in our two groups of 8:

$$E[U] = \frac{8 \times 8}{2} = 32, \qquad \mathrm{Var}(U) = \frac{8 \times 8 \times 17}{12} \approx 90.7$$

That's an SD of about 9.5. Our observed $U = 50$ is about 1.9 SDs above that expectation ($\frac{50-32}{9.5} \approx 1.9$).

R's $p = 0.063$ comes from this normal approximation, with the same two corrections as in the Wilcoxon case. The step correction raises p more than the tie adjustment lowers it, so p ends up slightly above what a z of 1.9 alone would give. For small samples without ties, R can instead read the exact null distribution directly, no approximation needed.

We simulate this null distribution the same way as before. We pool the 16 ranks, then 20,000 times pick 8 of them at random to call sodium and compute U. Each shuffle gives one U, the value we would see if group labels had nothing to do with bat activity. The critical values marked on the figure come from listing all 12,870 splits, which is the exact null distribution the simulation approximates.

Our observed $U = 50$ falls just short of the upper critical value of 50.5, consistent with R's $p = 0.063$: close to significant, but not quite.

The full picture. The t-test gave $p = 0.024$, a significant result. The Mann-Whitney-Wilcoxon test on the same 16 sites gave $p = 0.063$, not significant at $\alpha = 0.05$. The t-test's p-value depends on a normality assumption that both groups fail, so it can't be taken at face value. A violated assumption can push a p-value too low or too high, and the t-test alone gives no way to tell which. If the data had been roughly normal, the t-test would have been the better choice, because it has more power.

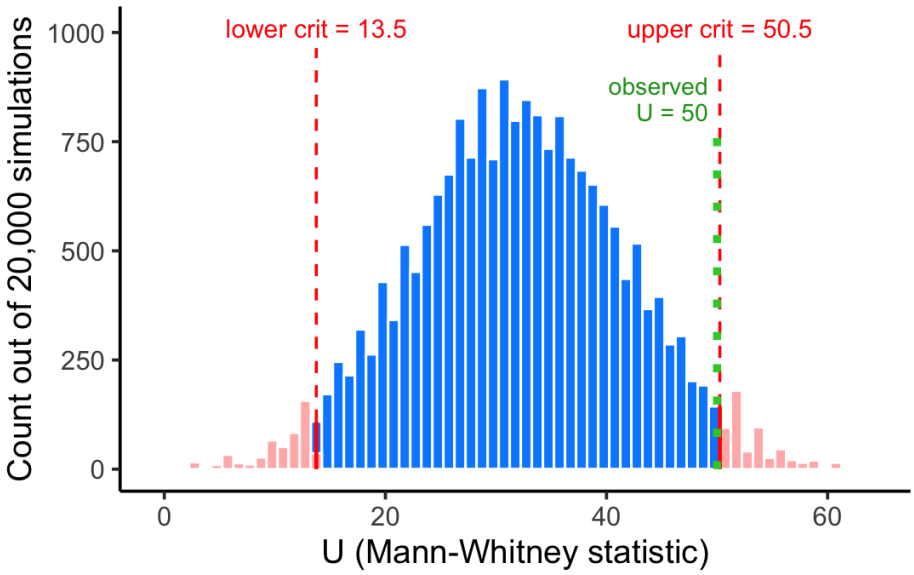


Figure 5: Simulated null distribution of the Mann-Whitney U statistic, built from 20,000 random reshuffles of group membership among the pooled, ranked counts of bat passes. Shaded regions mark the two-tailed 5% rejection region; our observed U = 50 falls just short of the upper critical value, 50.5, outside the rejection region.

Categorical data: the chi-square tests

Every test so far (the t-tests in Chapter 6, and the Wilcoxon and Mann-Whitney-Wilcoxon tests above) needs a continuous outcome variable. But a lot of environmental data isn't continuous at all: a site either has an invasive species or it doesn't; a stream sample either exceeds a safety threshold or it doesn't; a plot falls into one land-cover category, not a number line. None of the tests above apply to categorical data. Counts in categories need a different test, the chi-square (χ^2) test (Pearson 1900).

The test compares its statistic with the **chi-square distribution**. Because χ^2 adds up squared gaps between observed and expected counts, it can never be negative, and a mismatch in either direction makes it bigger. A χ^2 near 0 means the counts match the expectation closely, so only the upper tail counts against H_0. Like t, the chi-square distribution is a family of curves, one for each number of degrees of freedom (Figure 6). With few degrees of freedom it is sharply right-skewed, and as df grows it shifts right and becomes more symmetric.

Figure 7 shows what the rejection region looks like in practice, using the goodness-of-fit example we'll run below, which has $df = 3$. The critical value at $\alpha = 0.05$ is 7.81, and any statistic beyond that point (the shaded region) occurs less than 5% of the time by chance.

For the rank tests, the null distribution comes from every possible arrangement of the ranks under H_0, so it is exact. For the chi-square tests, the chi-square curve approximates the null distribution of the statistic, and the approximation is good only when expected counts are large enough. Each chi-square test below therefore comes with an expected-count rule.

There are two versions of the chi-square test, depending on the question you're asking.

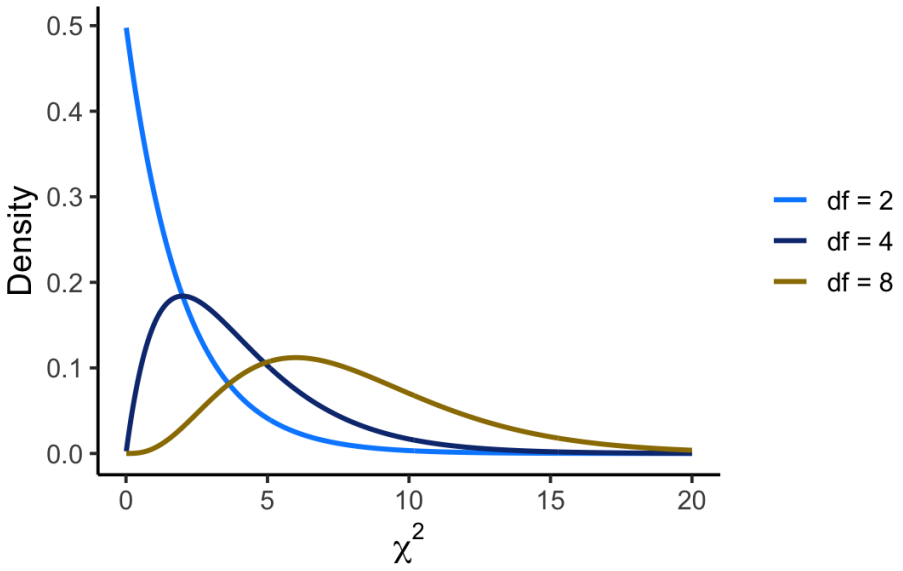


Figure 6: Chi-square distributions with 2, 4 and 8 degrees of freedom. Chi-square is never negative and is right-skewed. As df increases, the distribution shifts right and becomes more symmetric.

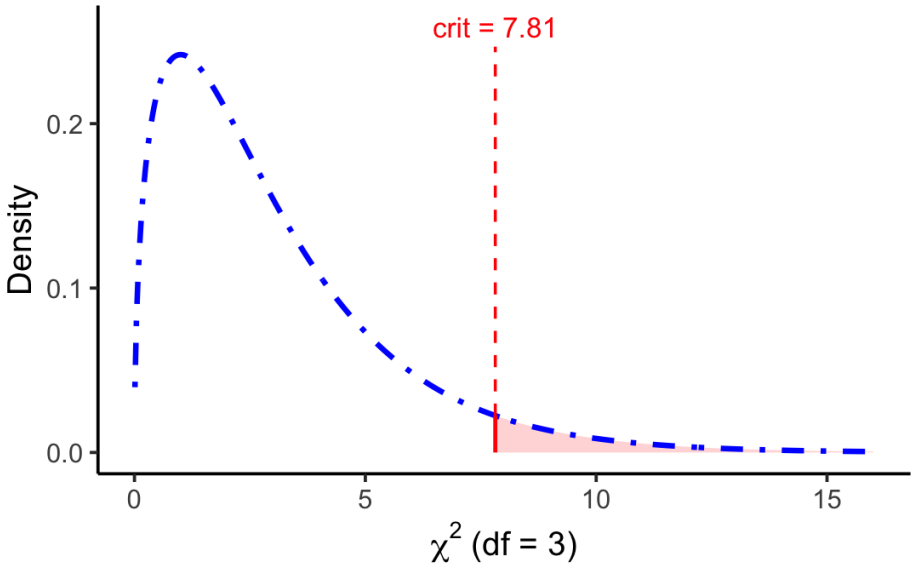


Figure 7: The chi-square distribution with df = 3. The shaded region marks the rejection region at $\alpha = 0.05$ (critical value 7.81).

The chi-square goodness-of-fit test

The test checks how well ("goodness") your observed counts match ("fit") a pattern you'd expect. The expectation can come from a historical baseline, a theory, or an assumption that every category is equally likely. A small chi-square statistic means a good fit (observed counts close to expected); a large one means a poor fit (observed counts far from expected).

A state wildlife agency has historically found fish evenly distributed across four zones of a lake, 25% in each. The counts in this example are invented. This year's survey of 200 fish found between 38 and 61 fish per zone. Figure 8 lays out the structure, which differs from every design so far in this book.

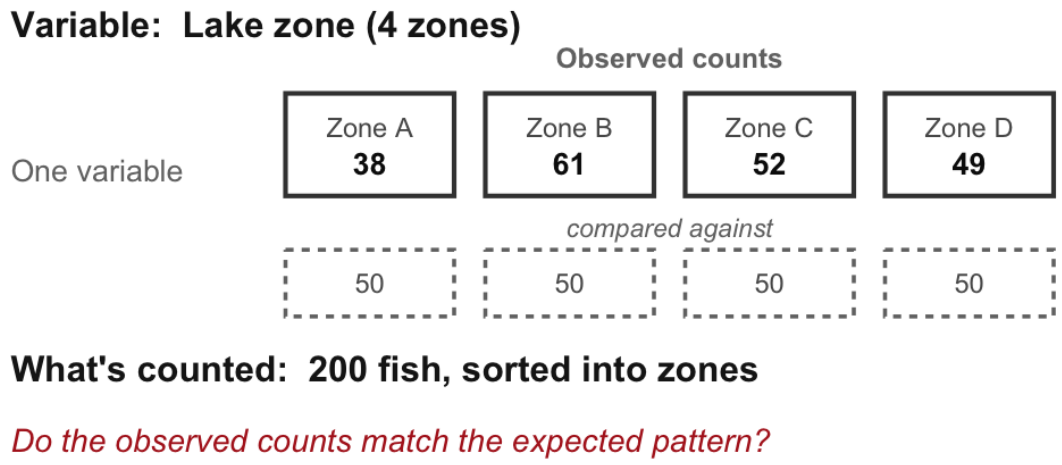


Figure 8: A goodness-of-fit design. There is no dependent variable measured on each unit: the counts themselves are the data, and they are compared against an expected pattern rather than against another group.

Every other test so far has had a dependent variable measured on each unit, usually compared across groups that an independent variable defines. Here there is only one variable, lake zone, and nothing is measured on the fish beyond which zone they turned up in. The counts *are* the data, and the comparison is against an expectation rather than against another group.

Table 12: Observed fish counts by lake zone (n = 200).

Zone	Observed
A	38
B	61
C	52
D	49

The question. Does this year's distribution across zones differ from the historical 25%-each pattern, or is this the kind of unevenness you'd expect from random sampling alone?

- H_0: fish are still distributed 25% per zone; the pattern matches the historical baseline.
- H_a: fish are not evenly distributed across zones; the pattern has changed.

```
chisq.test(observed_gof, p = rep(0.25, 4))
```

Table 13: Chi-square goodness-of-fit test.

Statistic (χ^2)	df	p
5.4	3	0.145

Assumption. The goodness-of-fit test makes no assumption about the shape of your data, since categories aren't numbers. It does require expected counts of about 5 or more in each category (Cochran 1954; Larntz 1978). The 5 is a conventional rule of thumb, not a value computed from these data or compared to a critical value. It checks whether the chi-square curve is a trustworthy approximation, and you check it before running the test.

Here, expected count per zone is $200 \times 0.25 = 50$ (total fish times the 25%-per-zone expectation under H_0), comfortably above that minimum. This is the expected-value idea from Chapter 3 applied to counts: the expected count in a category is the total times the probability H_0 gives that category. When expected counts are too small, the chi-square approximation breaks down and a different method (like Fisher's exact test) is more appropriate.

χ^2 adds one term per category. For each category, take the squared difference between observed and expected, scaled by expected: $\chi^2 = \sum \frac{(O-E)^2}{E}$.

Table 14: Chi-square contribution by zone.

Zone	Observed	Expected	Contribution
A	38	50	2.88
B	61	50	2.42
C	52	50	0.08
D	49	50	0.02

Summing the last column: $2.88 + 2.42 + 0.08 + 0.02 = 5.40$, matching R's statistic above. For goodness-of-fit, $df = k - 1$, where k is the number of categories: $4 - 1 = 3$. With $df = 3$, the critical value at $\alpha = 0.05$ is 7.81 (the same one marked on the distribution figure earlier in this chapter); our 5.40 falls well short of it.

Interpretation. $\chi^2(3) = 5.40$, $p = 0.145$, so the result is not significant. This year's zone counts are consistent with the historical 25%-each pattern. The unevenness we see is within the range chance alone would produce.

The chi-square test of independence

Now suppose researchers classify 150 city blocks by land use and by whether each block's peak summer surface temperature, measured from satellite images, topped 40°C. The data in this example are invented. Every block falls into exactly one cell of Figure 9.

In that table, nobody assigned anything. The blocks already had their land use and their summer temperatures before the researchers looked, and the cells simply count how many turned out each way. That makes this an observational design. Each block's temperature has been reduced to a yes or no, so the cells hold counts of blocks rather than temperatures.

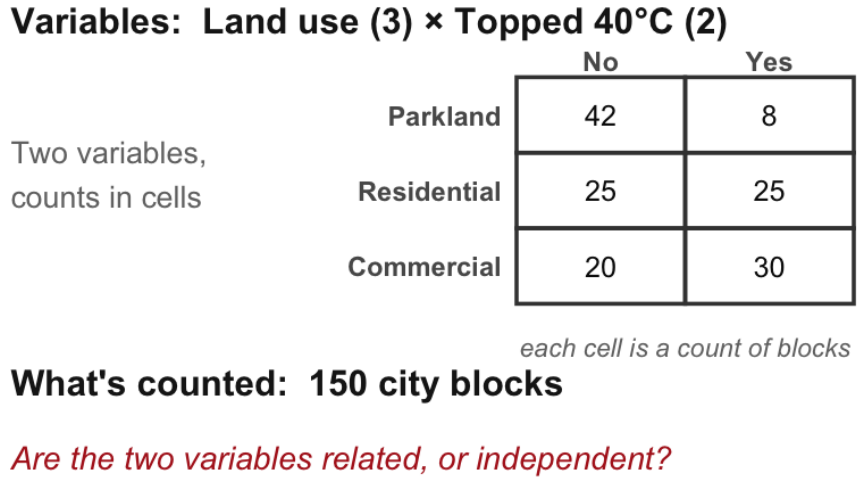


Figure 9: A test-of-independence design. Two categorical variables are crossed, but unlike a factorial experiment the cells hold counts of units that landed there, not conditions units were assigned to.

Table 15: City blocks by land use and whether summer surface temperature topped 40°C.

	No	Yes
Parkland	42	8
Residential	25	25
Commercial	20	30

The question. Is topping 40°C independent of land use, or are the two related?

- H_0: land use and topping 40°C are independent; knowing one tells you nothing about the other.
- H_a: land use and topping 40°C are related; they are not independent.

```
chisq.test(tab)
```

Table 16: Chi-square test of independence.

Statistic (χ^2)	df	p
21.84	2	< 0.001

Assumption. Same rule as goodness-of-fit: expected counts should generally be 5 or more in every cell of the table. Unlike the goodness-of-fit test, where expected counts came from a stated proportion (25% per zone), here they come from the table's own row and column totals: the expected count for a cell is $\frac{\text{row total} \times \text{column total}}{\text{grand total}}$, the count you'd see in that cell if land use and temperature really were unrelated. For Parkland/No, that's $\frac{50 \times 87}{150} = 29$ (50 parkland blocks total, 87 blocks total that did not top 40°C, 150 blocks overall).

Table 17: Expected counts under H0 (independence).

	No	Yes
Parkland	29	21
Residential	29	21
Commercial	29	21

All six expected counts are between 21 and 29, well above the 5-count minimum. This check does not tell us whether the test is significant. It only tells us whether the chi-square approximation is trustworthy here.

For a table, χ^2 adds one term per cell. The formula is the same as for goodness-of-fit, $\chi^2 = \sum \frac{(O-E)^2}{E}$, using the observed table and the expected table above.

Table 18: Chi-square contribution by cell.

	No	Yes
Parkland	5.83	8.05
Residential	0.55	0.76
Commercial	2.79	3.86

Summing all six cells: $5.83 + 8.05 + 0.55 + 0.76 + 2.79 + 3.86 = 21.84$, matching R's statistic above. For a test of independence, $df = (\text{rows} - 1)(\text{columns} - 1)$: $(3-1)(2-1) = 2$. The number of blocks never enters. With $df = 2$, the critical value at $\alpha = 0.05$ is 5.99; our 21.84 is more than three times that, which is where $p < 0.001$ comes from.

Interpretation. $\chi^2(2) = 21.84$, $p < 0.001$, so the result is significant. Topping 40°C is not independent of land use. To see how the two variables are related, look at the table. Divide within each row to get the share of blocks that topped 40°C for each land use (Table 19).

Table 19: Share of city blocks that topped 40°C, by land use.

Land use	Topped 40°C / total	Share
Parkland	8 / 50	16%
Residential	25 / 50	50%
Commercial	30 / 50	60%

Dividing within a row matters because raw counts only compare fairly when the row totals match. Residential and commercial blocks topped 40°C three to four times as often as parkland blocks. The test says the two variables are related, but, like correlation, it does not by itself establish that land use *causes* the heat.

Table 20 collects the tests in this chapter by the kind of data and design each one handles.

Table 20: Choosing a test in this chapter.

Outcome	Design	Parametric test (Chapter 6)	Test in this chapter
Continuous	One group vs. a reference, or paired (before/after)	One-sample or paired t-test	Wilcoxon signed-rank test; sign test if the differences are still not symmetric

Outcome	Design	Parametric test (Chapter 6)	Test in this chapter
Continuous	Two independent groups	Independent-samples t-test	Mann-Whitney-Wilcoxon test
Ordinal	Either design above	None	The same rank tests
Categorical (counts)	One categorical variable against an expected pattern	None	Chi-square goodness-of-fit test
Categorical (counts)	Two categorical variables	None	Chi-square test of independence

! Results reporting: nonparametric and chi-square tests

These tests report differently from a t-test. There is no mean difference to quote, because none of them used the means. Give the test's own statistic, the p-value, and, for the rank tests, a median rather than a mean.
Wilcoxon signed-rank, on the restoration differences:

> Soil moisture did not change significantly after restoration, Wilcoxon signed-rank $V = 39$, $p = 0.057$ (median change +1.5 percentage points across all 12 plots; 9 non-zero differences).

Mann-Whitney-Wilcoxon, on the bat activity comparison:

> Bat activity did not differ significantly between sites under sodium and LED streetlights, $U = 50$, $p = 0.063$ (medians 17 vs. 11.5 passes per night).

Chi-square, where the counts themselves are the result:

> Whether a city block topped 40°C was not independent of land use, $\chi^2(2) = 21.84$, $p < 0.001$. Parkland blocks topped 40°C in 8 of 50 cases, compared with 25 of 50 residential and 30 of 50 commercial blocks.

Report the number of non-zero differences for a signed-rank test, since dropped zeros change the effective sample size. For chi-square, give the counts as well as the statistic, because the counts are what a reader can interpret.

Chapter Summary

Why it matters. Real environmental data often fails the assumptions a t-test needs: small samples, skewed distributions, or outcomes that are counts and categories rather than measurements. This chapter covers the tests for those cases and how to tell when you need one. Chapter 8 compares three or more groups, with the Kruskal-Wallis test as its rank-based option.

Core ideas

- **Rank-based tests trade power for weaker assumptions.** The Wilcoxon signed-rank and Mann-Whitney-Wilcoxon tests ask the same questions as the paired and independent-samples t-tests, but work from ranks instead of raw magnitudes. That buys a milder distributional assumption at the cost of some sensitivity.
- **The sign test uses even less of the data.** It looks only at the direction of each difference. On the same restoration data, the t-test, Wilcoxon, and sign test gave p-values of 0.026, 0.057 and 0.508, because each one uses less of the data than the last.

- **Nonparametric tests have their own assumptions.** Wilcoxon assumes the differences are symmetric about their median. Mann-Whitney-Wilcoxon assumes the two distributions have the same *shape*, which is not the same as equal variance. State the assumption you are actually making.
- **Check assumptions before you trust a p-value, not after.** Both worked examples in this chapter ran the parametric test first, then went back and found the violation.
- **Categorical data needs a different tool entirely.** The chi-square tests handle counts: goodness-of-fit asks whether one variable's counts match an expected pattern, and the test of independence asks whether two categorical variables are related.
- **The testing logic never changes.** State H_0 and H_a, compute a statistic, and ask how surprising that statistic would be if H_0 were true. Only the statistic and its null distribution differ from test to test.

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